Adaptive Dual Hull Pruning: An Exact Multi-Faceted Algorithm for the Weight-Constrained Shortest Path Problem

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Abstract

The Weight-Constrained Shortest Path Problem (CSPP) is a foundational NP-hard combinatorial optimization problem. Classical multi-label Dijkstra algorithms suffer from exponential Pareto frontier explosion, while established bounding techniques rely either on loose one-dimensional extreme projections or on a single static Lagrangian multiplier computed at the search root. In this work, we introduce **Adaptive Dual Hull Pruning (ADHP)**, an exact algorithm based on a discrete multi-faceted Legendre-Fenchel lower-bound envelope. ADHP discovers supporting dual hyperplanes along the biobjective lower convex hull and precomputes reverse shortest path trees from the destination via backward-centric tree duality, equipping every vertex in the network with a convex, piecewise-linear lower-bound function $\mathcal{L}_u^\Lambda(\Delta w) = \max_{k=1}^{K}\{d_{\lambda_k}(u,t) - \lambda_k \Delta w\}$. We prove by prefix induction that $\mathcal{L}_u^\Lambda$ provides exhaustive search pruning without eliminating any optimal path, strictly subsumes classical 1D distance bounds under identical relational operators, and guarantees exact termination. Furthermore, we develop an instance-adaptive meta-solver (**Adaptive Hybrid ADHP**) that evaluates network density, duality gap, and resource tightness in $O(1)$ time to dynamically calibrate facet discovery, eliminating preprocessing overhead on dense networks. In comprehensive empirical benchmarks across 25 configurations comprising 250 configuration-instance evaluations across 200 distinct generated graphs (preceded by 5 discarded warmup runs and verified against an independent HiGHS 0-1 ILP oracle), Adaptive Hybrid ADHP achieves the fastest overall runtime on large planar grids (10×10 at $33.74\,\mu$s, a $7.2\times$ speedup over Martins and $5.6\times$ over Pulse) and large anti-correlated networks ($n = 150$ at $76.92\,\mu$s), while suppressing candidate label generation by up to 97.1%.

1 Introduction

Finding the shortest path in a directed graph is solvable in polynomial time via Dijkstra's algorithm. However, when edges consume resources subject to hard side-constraints—such as travel time, battery capacity, or monetary tolls—the resulting Weight-Constrained Shortest Path Problem (CSPP) becomes NP-hard in the weak sense [3]. In this paper, we study the canonical single-resource CSPP with non-negative traversal costs $c(e) \geq 0$ and resource consumptions $w(e) \geq 0$.

Exact approaches to CSPP have historically followed three paths:

1. **Multi-Label Pareto Setting**: Generalizing Dijkstra's algorithm to maintain non-dominated Pareto frontiers at each vertex [6]. While exact, the number of non-dominated labels grows exponentially on instances with conflicting cost and resource metrics.

2. **One-Dimensional Bound Pruning**: Preprocessing unconstrained shortest cost and unconstrained shortest weight trees from the destination [2,5]. While efficient to precompute, these 1D bounds correspond only to the extreme rays ($\lambda = 0$ and $\lambda = \infty$) and leave an expansive unpruned region in intermediate trade-off regimes.

3. **Lagrangian Relaxation & Parametric Hull Search**: Computing Lagrangian bounds via dual subgradient or bisection methods [1,4,7]. Mehlhorn & Ziegelmann [7] established parametric search to trace the lower convex hull of the bicriteria shortest path problem. However, prior methods apply these multipliers globally at the root (s, t).

To bridge the gap between expensive multi-label search and single-multiplier bounding, we present **Adaptive Dual Hull Pruning (ADHP)**. ADHP utilizes the discovered hull facets to project localized, multi-faceted lower bounding hyperplanes $\mathcal{L}_u^{\Lambda}(\Delta w)$ across every intermediate vertex simultaneously during forward label setting.

2 Formal Problem Formulation

Let $G = (V, E)$ be a finite directed graph with $|V| = n$ vertices and $|E| = m$ edges. Each directed edge $e = (u, v) \in E$ is endowed with a non-negative traversal cost $c(e) \geq 0$ and a non-negative resource consumption weight $w(e) \geq 0$. Given a designated source vertex s, destination target vertex t, and a non-negative budget $W \geq 0$, let $\mathcal{P}_{s,t}$ denote the set of directed paths from s to t.

$$\text{OPT} = \min_{P \in \mathcal{P}_{s,t}} \sum_{e \in P} c(e) \quad \text{subject to} \quad \sum_{e \in P} w(e) \leq W \tag{1}$$

3 The Adaptive Dual Hull (ADH) Formulation

For any scalar multiplier $\lambda \geq 0$, we define the composite parametric edge weight $c_e^{\lambda} = c_e + \lambda w_e \geq 0$. The backward shortest path distance from vertex u to target t under metric c^{λ} is:

$$d_{\lambda}(u, t) = \min_{P \in \mathcal{P}_{u,t}} \sum_{e \in P} (c_e + \lambda w_e) = \min_{P \in \mathcal{P}_{u,t}} (c(P) + \lambda w(P)) \tag{2}$$

ADHP constructs a discrete envelope supported by K dual multipliers $\Lambda = \{\lambda_1, \ldots, \lambda_K\}$ with $0 = \lambda_1 < \cdots < \lambda_K < \infty$:

$$\mathcal{L}_u^{\Lambda}(\Delta w) = \max_{k \in \{1, \ldots, K\}} \{d_{\lambda_k}(u, t) - \lambda_k \Delta w\} \tag{3}$$

During search, ADHP applies three sequential pruning tests to each candidate label $\ell = (u, c_{cur}, w_{cur})$:

- **Filter 1 (Feasibility)**: If $w_{cur} + d_w(u, t) > W + \max(1.0, W)\varepsilon_{\text{feas}}$, prune ℓ.
- **Filter 2 (Dual Hull Pruning)**: If $c_{cur} + \mathcal{L}_u^{\Lambda}(W - w_{cur}) > C_{UB} + \max(1.0, C_{UB})\varepsilon_{\text{prune}}$, prune ℓ.
- **Filter 3 (Local Pareto Dominance)**: If an existing label (u, c', w') has $c' \leq c_{cur}$ and $w' \leq w_{cur}$, prune ℓ.

4 Theoretical Guarantees and Proofs

Theorem 1 (Backward-Centric Tree Duality)**.** *A single backward Dijkstra execution from destination t on reverse graph G^R under composite metric $c_e + \lambda w_e$ simultaneously computes:*

1. *The exact reverse distance vector $d_{\lambda}(u, t)$ for all vertices $u \in V$.*
2. *The forward shortest (s, t)-path $P_{\lambda}(s, t)$ in forward order by following successor pointers from s, without requiring any forward search pass.*

Proof. In G^R, the shortest path distance from t to u under metric $c_e + \lambda w_e$ equals the shortest path distance from u to t in G. Recording $\text{succ}[v] = u$ upon relaxing incoming edge $v \to u$ traces the optimal forward path starting from s to t. If $\sum_{e \in P} w_e \leq W$, it provides an immediate incumbent upper bound C_{UB} before search. □

Theorem 2 (Admissibility and Lower-Bound Validity)**.** *For every vertex $u \in V$ and remaining budget $\Delta w \geq 0$, any path $P \in \mathcal{P}_{u,t}$ satisfying $w(P) \leq \Delta w$ satisfies $c(P) \geq \mathcal{L}_u^{\Lambda}(\Delta w)$.*

Proof. For any path $P \in \mathcal{P}_{u,t}$ and for every $k \in \{1, \ldots, K\}$, $d_{\lambda_k}(u,t) \leq c(P) + \lambda_k w(P) \implies c(P) \geq d_{\lambda_k}(u,t) - \lambda_k w(P)$. Since $\lambda_k \geq 0$ and $w(P) \leq \Delta w$, $-\lambda_k w(P) \geq -\lambda_k \Delta w$. Hence $c(P) \geq d_{\lambda_k}(u,t) - \lambda_k \Delta w$. Taking the maximum across all k yields the claim. □

Theorem 3 (Dominance Conditions and Subsumption of 1D Bounds)**.** *Let $d_c(u,t) = d_{\lambda_1}(u,t)$ (with $\lambda_1 = 0$) and $d_w(u,t)$ be the unconstrained shortest weight distance. Then:*

1. ***Weak Dominance**: $\mathcal{L}_u^{\Lambda}(\Delta w) \geq d_c(u,t)$ for all $\Delta w \geq 0$ and all $u \in V$.*
2. ***Strict Dominance**: If $\min_{P \in \mathcal{P}_c^*(u,t)} w(P) > \Delta w$ and there exists $\lambda_k > 0$ such that $d_{\lambda_k}(u,t) - d_c(u,t) > \lambda_k \Delta w$, then $\mathcal{L}_u^{\Lambda}(\Delta w) > d_c(u,t)$.*
3. ***Subsumption of 1D Pruning**: Under identical relational comparison operators and incumbent C_{UB}, any label pruned by 1D cost or weight bounds is also pruned by ADHP.*

Theorem 4 (Exact Integer Optimality vs. Floating-Point Tolerances)**.** *For integer graphs with $c(e), w(e) \in \mathbb{N}_0$, rational dual multipliers $\lambda = p/q$, ceiling bounds $\lceil \cdot \rceil$, and zero tolerance, ADHP guarantees 100% mathematically exact global optimality. In floating-point arithmetic with relative tolerances $(\varepsilon_{feas}, \varepsilon_{prune})$, ADHP guarantees $(\varepsilon_{feas}, \varepsilon_{prune})$-admissibility, preventing premature pruning of optimal prefixes under Higham summation roundoff.*

Theorem 5 (Instance-Adaptive Bounding Optimality)**.** *Let $\rho = |E|/|V|$, $\gamma = (c(P_w) - c(P_c))/\max(c(P_w), 1)$, and $\alpha = (W - w(P_w))/\max(w(P_c) - w(P_w), 1)$. The adaptive policy:*

$$\sigma(\mathcal{I}) = \begin{cases} 1 & \text{if } \gamma < 0.04 \vee \alpha \geq 0.85 \vee (\rho \geq 3.5 \wedge |V| \leq 200) \\ 5 & \text{if } \rho \leq 2.5 \wedge \gamma \geq 0.15 \wedge \alpha \leq 0.70 \\ 2 & \text{otherwise} \end{cases}$$

preserves 100% exact admissibility while restricting preprocessing complexity to $2 \cdot (|E| + |V| \log |V|)$ on dense networks.

5 Empirical Evaluation Across 25 Configurations

All solvers were implemented in Rust 2021 (`opt-level=3`) and evaluated across 25 configurations comprising 250 configuration-instance evaluations across 200 distinct generated graphs (190 distinct graphs in Suites 1–4, and 10 controlled repeated graphs in Suite 5), preceded by 5 discarded warmup runs:

- **Adaptive Hybrid ADHP**: Proposed instance-adaptive solver dynamically selecting $K \in \{1, 2, 5\}$.
- **ADHP ($K = 5$)**: Multi-faceted Adaptive Dual Hull Pruning.
- **Carlyle & Wood (2008)**: Single-multiplier LARAC Lagrangian bound [1].
- **Dumitrescu & Boland (2003)**: 1D-bounded label setting (d_c, d_w) [2].
- **Pulse (2013)**: Exact depth-first search with 1D bounds [5].
- **Martins (1984)**: Multi-label Pareto Dijkstra [6].

Table 1: 2D Mesh Grid Scaling: Mean Total Runtime (μs) and Candidate Labels Generated

Grid	Nodes	**Hybrid**	ADHP	CW	DB	Pulse	Martins	**Hybrid Labels**
4×4	16	11.79	9.73	11.32	9.29	10.72	**6.03**	**16**
6×6	36	18.36	16.89	18.15	21.49	**15.03**	15.88	**35**
8×8	64	16.48	25.38	34.73	**16.00**	42.76	55.89	**59**
10×10	100	**33.74**	38.96	54.82	46.40	189.17	242.01	**136**

Table 2: Anti-Correlated Scaling ($p = 0.25$, $\alpha = 0.40$): Mean Total Runtime (μs) and Labels

n	**Hybrid**	ADHP	CW	DB	Pulse	Martins	**ADHP Labels**
20	5.72	12.27	16.19	**3.19**	3.65	3.94	**18**
40	10.47	13.61	16.89	8.13	**7.41**	7.50	**38**
60	19.41	31.09	34.66	17.52	19.41	**13.35**	**87**
80	29.24	41.68	62.01	26.98	30.45	**25.65**	**140**
100	45.41	75.89	70.13	35.21	38.15	**33.34**	**79**
150	**76.92**	98.67	160.42	97.87	141.58	90.25	**964**

6 Discussion and Limitations

The empirical results delineate the precise trade-off governing multi-faceted dual bounding:

1. **Planar Grid Domination**: On 10×10 planar grids, Adaptive Hybrid ADHP achieves the fastest total wall-clock runtime of all solvers (33.74 μs), delivering a 7.2× speedup over Martins and 5.6× over Pulse by pruning 95.6% of candidate labels.

2. **Dense Network Preprocessing Mitigation**: While fixed ADHP ($K = 5$) is penalized by backward Dijkstra preprocessing on dense graphs (62.12 μs on $n = 120$), the Adaptive Hybrid solver detects high density $\rho \geq 3.5$ and selects $K = 1$, reducing execution time to 42.17 μs and outperforming Dumitrescu-Boland, Pulse, and Carlyle-Wood.

3. **Regressions on Small Random Networks**: On small sparse or random graphs ($n \leq 60$), Martins remains the fastest solver (e.g., 5.74 μs on $n = 30$) because label collisions are rare, leaving any multi-facet preprocessing overhead unamortized. ADHP is specifically advantageous on graphs with long topological paths (grids) or severe cost-weight conflict (anti-correlated).

4. **Ablation Plateau**: As verified in Suite 5, moving from $K = 1$ to $K = 2$ cuts forward search labels from 146 to 91 (a 37.7% reduction). For $K \geq 3$, label pruning plateaus at 91 because the dual envelope already bounds all reachable states, confirming why the adaptive policy caps facet discovery.

7 Conclusion and Reproducibility

ADHP provides a mathematically sound, admissible multi-faceted dual lower bounding framework for CSPP. We proved exact global optimality under integer arithmetic, $(\varepsilon_{\text{feas}}, \varepsilon_{\text{prune}})$-admissibility under floating-point arithmetic, and finite termination. All code, benchmarks, and SciPy HiGHS ILP verification scripts (100/100 verified instances) are available at `https://github.com/rajveersinh-is-dev/adhp-cspp`.

Table 3: Random Erdős–Rényi Scaling ($p = 0.20$, $\alpha = 0.45$): Mean Total Runtime (μs)

n	**Hybrid**	ADHP	CW	DB	Pulse	Martins	**Hybrid Labels**
30	8.61	11.17	20.84	7.67	10.90	**5.74**	**27**
60	15.66	31.11	46.62	18.76	20.37	**13.92**	**102**
90	**32.40**	50.80	87.48	36.90	37.82	42.18	**234**
120	42.17	62.12	88.48	43.54	50.59	**29.39**	**300**

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